

Mphil Thesis Defense

Yang Hu

HKUST

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- 4 Enriched ∞ -category theory
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- 7 A lax symmetric monoidal $(\infty, 2)$ functor $\text{Span}_2(\mathcal{C}) \rightarrow 2\text{Pr}_{\kappa}$
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Main Conclusion

Given a sheaf theory, which is a lax symmetric monoidal functor:

$$\mathrm{Shv}(-) : \mathrm{Span}_1(\mathcal{C})^{\otimes} \rightarrow \mathrm{Pr}^{L, \otimes}$$

, where $\mathcal{C}$ is a category of geometric objects with finite limits. Then we can construct a lax symmetric monoidal $(\infty, 2)$ functor:

$$\mathrm{Span}_2(\mathcal{C})^{\otimes} \rightarrow (\mathrm{Cat}^{\mathrm{Pr}^L})^{\otimes}$$

. Especially if the input sheaf theory lands in Pr_{κ}^L for some regular cardinality κ , then our $2\mathrm{Shv}(-)$ can be taken as a lax symmetric $(\infty, 2)$ functor:

$$2\mathrm{Shv}(-) : \mathrm{Span}_2(\mathcal{C}) \rightarrow 2\mathrm{Pr}_{\kappa}^L := \mathrm{Mod}_{\mathrm{Pr}_{\kappa}}(\mathrm{Pr}_{\kappa})$$

Main Ingredients

First of all, let me answer the following questions about the main characters appearing in the construction:

- What is the symmetric monoidal $(\infty, 2)$ category $\mathrm{Span}_2(\mathcal{C})$?
- What is a lax symmetric monoidal $(\infty, 2)$ functor?
- What is the symmetric monoidal $(\infty, 2)$ structure on $\mathrm{Cat}^{\mathrm{Pr}^L}$, and further, what is the symmetric monoidal $(\infty, 2)$ structure on $2\mathrm{Pr}_\kappa$?

And then let me explain:

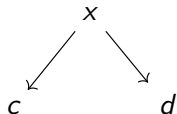
- What does this $2\mathrm{Shv}(-)$ look like?
- Why do we need a ‘higher sheaf’?
- Why do we need such a lax symmetric monoidal $(\infty, 2)$ functor?

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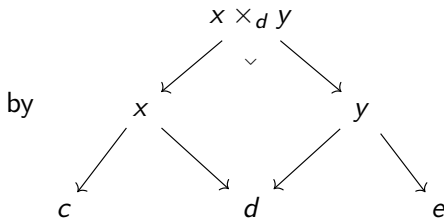
$\text{Span}_1(\mathcal{C})$

Given a category $\mathcal{C}$ with finite limits, the category of span $\text{Span}_1(\mathcal{C})$ has:

- Object: $c \in \mathcal{C}$
- Morphism: $f : c \rightarrow d \in \text{Span}_1(\mathcal{C})$ is a diagram



- And the composition of morphism $f : c \rightarrow d$ with $g : d \rightarrow e$ is given



$\text{Span}_2(\mathcal{C})$ and $\text{Span}_{1\frac{1}{2}}(\mathcal{C})$

We can enhance $\text{Span}_1(\mathcal{C})$ into a 2 category with 2 morphism between 1

morphisms $\begin{array}{ccc} & x & \\ \swarrow & & \searrow \\ c & & d \end{array}$ and $\begin{array}{ccc} & y & \\ \swarrow & & \searrow \\ c & & d \end{array}$ as the following

diagram: $\begin{array}{ccccc} & & x & & \\ & \swarrow & \uparrow & \searrow & \\ c & \leftarrow & z & \rightarrow & d \\ & \nwarrow & \downarrow & \nearrow & \\ & & y & & \end{array} .$

Another enhancement of $\text{Span}_1(\mathcal{C})$, denoted as $\text{Span}_{1\frac{1}{2}}(\mathcal{C})$, where the 2 morphism above satisfies $x \cong z$

Sheaf theory

A sheaf theory $\mathrm{Shv}(-)$ can be encoded into a lax symmetric monoidal functor

$$\mathrm{Shv}(-) : \mathrm{Span}_1(\mathcal{C}) \rightarrow \mathrm{Pr}^L$$

, it sends $x \in \mathcal{C}$ to $\mathrm{Shv}(x)$, and given a morphism $c \xleftarrow{f} x \xrightarrow{g} d$, the left leg is sent to $f^* : \mathrm{Shv}(c) \rightarrow \mathrm{Shv}(x)$, and the right leg is sent to $g_! : \mathrm{Shv}(x) \rightarrow \mathrm{Shv}(d)$. The lax symmetric monoidal structure enables us to define the operation $\otimes : \mathrm{Shv}(x) \otimes \mathrm{Shv}(x) \rightarrow \mathrm{Shv}(x)$: for any $x \in \mathrm{Span}_1(\mathcal{C})$, it's a commutative algebra object, with the multiplication given by $x \times x \xleftarrow{\Delta} x = x$, and unit given by $* \leftarrow x = x$, and thus $\mathrm{Shv}(x)$ is a commutative algebra object in Pr^L , and thus a presentably symmetric monoidal category. Moreover, such a functor encodes the symmetric monoidality of $*$ pullback, base change, and the projection formula. By taking right adjoints of $*$, $!$ and $\otimes$, we get in total six operations $*$ $\dashv$ $*$, $!$ $\dashv$ $!$, $\otimes \dashv \mathcal{H}om$ on $\mathrm{Shv}(-)$.

Rigorous Formulation I: n-Segal space

Intuition

An ordinary category is a set of objects **Ob** and a set of morphisms **Mor**, with

- Source and target maps $s, t : \mathbf{Mor} \rightarrow \mathbf{Ob}$,
- And the composition is encoded into a map $comp : \mathbf{Mor} \times_{s \circ t} \mathbf{Mor} \rightarrow \mathbf{Mor}$ satisfying compatibility conditions

$$\begin{array}{ccc} \mathbf{Mor} \times_{\mathbf{Ob}} \mathbf{Mor} \times_{\mathbf{Ob}} \mathbf{Mor} & \xrightarrow{id \times comp} & \mathbf{Mor} \times_{\mathbf{Ob}} \mathbf{Mor} \\ \downarrow comp \times id & & \downarrow comp \\ \mathbf{Mor} \times_{\mathbf{Ob}} \mathbf{Mor} & \xrightarrow{comp} & \mathbf{Mor} \end{array}$$

- The identity map $e : \mathbf{Ob} \rightarrow \mathbf{Mor}$, with compatibility:

$$\begin{array}{ccc} \mathbf{Ob} \times_{\mathbf{Ob}} \mathbf{Mor} & \xrightarrow{\cong} & \mathbf{Mor} \\ \downarrow e \times id & & \parallel \\ \mathbf{Mor} \times_{\mathbf{Ob}} \mathbf{Mor} & \xrightarrow{comp} & \mathbf{Mor} \end{array}$$

Rigorous Formulation: n -Segal space

For the higher category, we need more data to organize the coherence. Let Δ be the simplex category:

Definition

An 1-fold Segal space is a functor $F : \Delta^{op} \rightarrow \mathbf{Spc}$ satisfying the Segal condition:

$$F([n]) \cong F([1]) \times_{F([0])} \cdots \times_{F([0])} F([1])$$

Definition

An n -fold Segal space is a functor $F : (\Delta^{op})^n \rightarrow \mathbf{Spc}$ satisfying the Segal condition:

- F satisfies the n -fold Segal condition.
- $F([0, \bullet]) : (\Delta^{op})^{n-1} \rightarrow \mathbf{Spc}$ is constant. (This is the underlying space of objects)
- $F([k, \bullet]) : (\Delta^{op})^{n-1} \rightarrow \mathbf{Spc}$ is an $(n-1)$ -fold Segal space for $k > 1$.

Rigorous Formulation: n -Segal space

Remark

By definition, an n -fold Segal space consists of

- space of objects $F([0, 0, \dots, 0])$,
- space of 1-morphisms $F([1, 0, 0, \dots, 0])$,
- space of 2-morphisms $F([1, 1, 0, \dots])$,
- ...
- space of n -morphisms $F([1, 1, \dots, 1])$.

Given $x, y \in F([0, 0, \dots, 0])$, the space of 1-morphisms $\text{Mor}(x, y)$ can be

$$\begin{array}{ccc} \text{Mor}(x, y) & \longrightarrow & F([1, 0, \dots, 0]) \\ \downarrow \lrcorner & & \downarrow s \times t \\ * & \xrightarrow[\{(x, y)\}]{} & F([0, 0, \dots])^{\times 2} \end{array}, \text{ and}$$

computed as the fiber product

given $f, g \in \text{Mor}(x, y)$, we can compute the space of 2-morphisms between f, g as $F([1, 1, 0, \dots, 0]) \times_{F([1, 0, \dots, 0])^{\times 2}} *$, where $*$ specifies the pair (f, g) .

Rigorous Formulation: univalent/complete n -Segal space

Let $\text{Seg}_n(\text{Spc})$ denote the category of n -fold Segal space. Let E^n denote the groupoid with n objects and a unique morphism between each pair of objects; then through the embedding $\text{Cat}_1 \cong \text{Seg}_1(\text{Set}) \rightarrow \text{Seg}_1(\text{Spc})$, we can regard E^n as an object in $\text{Seg}_1(\text{Spc})$.

Definition

A segal space X is univalent/complete if $X_0 \rightarrow \text{Map}_{\text{Seg}_1(\text{Spc})}(E^1, X)$ is an equivalence.

Inductively, we have:

Definition

An n -fold Segal space X is univalent/complete if $X_{\bullet,0,\dots,0}$ is complete Segal space, and for any $x, y \in X_{0,\dots,0}$, $\text{Mor}_X(x, y) \cong X_{1,\bullet} \times_{X_{0,\bullet}^{\times 2}} \{x, y\}$ is a complete $(n-1)$ -fold Segal space.

By definition, the full subcategory of complete n -fold Segal space is a localization of $\text{Seg}_n(\text{Spc})$ with respect to morphisms $E^1 \rightarrow E^0$, and serves as a model of (∞, n) categories, denoted as Cat_n .

Example

Consider the $(1,1)$ category of ordinary categories $(1,1) \text{Cat} \cong \text{Seg}_1(\text{Set})$, then $(1,1) \text{Cat} \hookrightarrow \text{Seg}_1(\text{Spc}) \xrightarrow{\text{localization}} \text{Cat}_1$ induces a localization $(1,1) \text{Cat} \rightarrow (2,1) \text{Cat}$ where the latter is the $(2,1)$ category of ordinary categories by inverting the fully faithful and essential surjective functors.

Example

Let S^1 denote the complete Segal space defined by $\Delta^{op} \rightarrow \text{Spc}, [n] \mapsto \text{Map}_{\text{Spc}}([n], S^1)$, then S^1 is the localization of $B\mathbf{Z}$ in Cat_1 .

$\text{Span}_n(\mathcal{C})$ by Hugseng

Construction

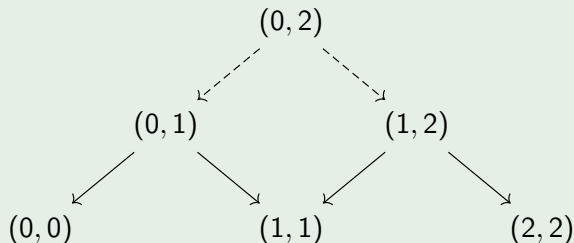
Given a category with finite limits, consider the following n -fold simplicial space:

$$\overline{\text{Span}}_n(\mathcal{C}) : (\Delta^{op})^{\times n} \rightarrow \text{Spc}$$

sending $[k_1, \dots, k_n]$ to $\text{Map}_{\text{Cat}}(\text{Tw}(k_1) \times \dots \times \text{Tw}(k_n), \mathcal{C})$

Example

$\text{Tw}([1]): (0,0) \leftarrow (0,1) \rightarrow (1,1)$; $\text{Tw}([2]):$



Let $n = 1$, and $\Lambda_k \hookrightarrow \text{Tw}([k])$ be the full subcategory of objects (i, j) such that $j - i \leq 1$.

Definition

A functor $F : \text{Tw}([k]) \rightarrow \mathcal{C}$ is Cartesian iff the natural map $F \rightarrow \text{RKan}_{\Lambda_k \hookrightarrow \text{Tw}([k])}(F|_{\Lambda_k})$ is an equivalence.

Definition

A functor $F : \text{Tw}([k_1]) \times \cdots \times \text{Tw}([k_n]) \rightarrow \mathcal{C}$ is Cartesian if its restriction to any component $F|_{\{i_1\} \times \cdots \times \text{Tw}([k_j]) \times \{i_n\}}$ is Cartesian.

Construction

Let $\text{Span}'_n(\mathcal{C})$ be the full sub multi-simplicial space of $\text{Span}_n(\mathcal{C})$ consisting of Cartesian functors:

$$\text{Span}'_n(\mathcal{C})([n]) := \text{Map}_{\text{Cat}}^{\text{cart}}(\text{Tw}([k_1]) \times \cdots \times \text{Tw}([k_n]), \mathcal{C})$$

Theorem

(Haugsgeng) $\text{Span}'_n(\mathcal{C})$ satisfies the n -fold Segal condition.

However, it's not an n -fold Segal space yet; we further extract a subsimplicial object out of it in an inductive way:

Construction

For $n = 2$, let $\text{Span}_2(\mathcal{C}) \hookrightarrow \text{Span}_2(\mathcal{C})' : \Delta^{op} \rightarrow \text{Fun}(\Delta^{op}, \text{Spc})$ be:

$$\begin{array}{ccc} \text{Span}_2(\mathcal{C})([n], \bullet) & \longrightarrow & \text{Span}_2(\mathcal{C})'([n], \bullet) \\ \downarrow \lrcorner & & \downarrow \\ \text{Span}_2(\mathcal{C})'([0], [0])^{\times(n+1)} & \longrightarrow & \text{Span}_2(\mathcal{C})'([0], \bullet)^{\times(n+1)} \end{array} \quad \text{for } [n] \in \Delta^{op}.$$

Theorem

(Haugsgeng) $\text{Span}_n(\mathcal{C})$ is univalent.

Symmetric monoidal (∞, n) structure

The above construction of $\text{Span}_n(\mathcal{C})$ is functorial, thus we get a functor:

$$\text{Span}_n(-) : \text{Cat}^{pb} \rightarrow \text{Cat}_n$$

, which preserves finite products, thus it can be enhanced into a symmetric monoidal functor:

$$\text{Span}_n(-) : (\text{Cat}^{pb})^\times \rightarrow (\text{Cat}_n)^\times$$

, which induces a functor:

$$\text{Span}_n(-) : \text{CAlg}(\text{Cat}^{pb}) \rightarrow \text{CAlg}(\text{Cat}_n)$$

Definition

Symmetric monoidal (∞, n) category is an object in $\text{CAlg}(\text{Cat}_n)$

Theorem

For any $\mathcal{C}$ with finite limits, $\text{Span}_n(\mathcal{C}) \in \text{Cat}_n$ can be enhanced into a symmetric monoidal (∞, n) category.

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Lax symmetric monoidal $(\infty, 2)$ functor

Definition

A 2 functor $\pi : C \rightarrow D$ is a 2-coCartesian fibration if it satisfies the condition:

- (coCartesian 1 morphism) Given any $c_1 \in C$, and 1 morphism $e : \pi(c_1) \rightarrow d$ in D , there exists a 1 morphism $f : c_1 \rightarrow c_2$ such that $\pi(f) = e$ and f induces a pullback

diagram:

$$\begin{array}{ccc} \underline{\text{Map}}_C(c_2, x) & \xrightarrow{\quad} & \underline{\text{Map}}_C(c_1, x) \\ \downarrow & \lrcorner & \downarrow \\ \underline{\text{Map}}_D(\pi(c_2), \pi(x)) & \xrightarrow{\quad} & \underline{\text{Map}}_D(\pi(c_1), \pi(x)) \end{array}$$

for any

$x \in C$.

- (Cartesian 2 morphism) $\underline{\text{Map}}_C(x, y) \rightarrow \underline{\text{Map}}_D(\pi(x), \pi(y))$ is a Cartesian fibration.
- The composition map $\underline{\text{Map}}_C(x, y) \times \underline{\text{Map}}_C(y, z) \rightarrow \underline{\text{Map}}_C(x, z)$ preserves Cartesian edges.

By Joost Nuiten, there is an (∞, n) version of (Un)straightening : given a symmetric monoidal $(\infty, 2)$ category $M : \text{Comm}^{\otimes} \rightarrow \text{Cat}_2$, its unstraightening gives a 2 coCartesian fibration $M^{\otimes} \rightarrow \text{Comm}^{\otimes}$

Definition

A functor F between symmetric monoidal $(\infty, 2)$ categories $M^{\otimes}$ and $N^{\otimes}$ over $\text{Comm}^{\otimes}$ is a lax symmetric monoidal $(\infty, 2)$ functor if it satisfies the condition that

- F preserves 1 coCartesian edges over the inert maps in $\text{Comm}^{\otimes}$.
- F preserves 2 Cartesian edges.

Definition

A lax symmetric monoidal $(\infty, 2)$ functor is a symmetric monoidal $(\infty, 2)$ functor if it preserves all Cartesian edges over $\text{Comm}^{\otimes}$.

Remark

By Abellán-Gagna-Haugsgeng, a lax symmetric monoidal $(\infty, 2)$ functor corresponds to a lax natural transformation between $M : \text{Comm}^{\otimes} \rightarrow \text{Cat}_2$ and $N : \text{Comm}^{\otimes} \rightarrow \text{Cat}_2$.

Lax symmetric monoidal $(\infty, 2)$ functor

Symmetric monoidal $(\infty, 2)$ functor is easier to produce, they appear as a 1 morphism in $\text{CAlg}(\text{Cat}_2)$. And by the following theorem, we can produce lax symmetric monoidal $(\infty, 2)$ functor by taking right adjoint:

Definition

A pair of functors $f : C \rightleftarrows D : g$ in Cat_2 is called adjunction iff there exists natural transformation $\eta : \text{id} \Rightarrow gf$, and $\epsilon : fg \Rightarrow \text{id}$ satisfying the triangle identities.

Theorem

Given an adjunction $f : C \rightleftarrows D : g$ in Cat_2 , if f can be enhanced into a symmetric monoidal $(\infty, 2)$ functor $F : C^{\otimes} \rightarrow D^{\otimes}$, then g can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor.

In our construction, the lax symmetric monoidal $(\infty, 2)$ functors appear as the right adjoint of a symmetric monoidal $(\infty, 2)$ functor

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Enriched ∞ category: Haugseng's model of categorical algebra

Given a space S , let Ass_S be the coCartesian fibration classifying the functor:

$$\Delta^{op} \rightarrow \text{Set}^{op} \xrightarrow{\text{Fun}(-, S)} \text{Cat}$$

Definition

(Haugsgeng-Gepner) Given a monoidal category $\mathcal{V}^\otimes$, a category algebra in $\mathcal{V}$ with underlying space is a morphism over Δ^{op}

$$F : \text{Ass}_S \rightarrow \mathcal{V}^\otimes$$

preserving the inert liftings over Δ^{op} .

Remark

The restriction of F to the fiber $[1] \in \Delta^{op}$ is the enriched Hom functor: $S \times S \rightarrow \mathcal{V}$, sending $(x, y) \in S \times S$ to $\underline{\text{Hom}}(x, y) \in \mathcal{V}$.

Enriched ∞ category

We can assemble the categorical algebras in $\mathcal{V}$ with different underlying spaces together into an ∞ category: Let $\text{Alg}_{\text{cat}}(\mathcal{V})$ be the pullback

$$\begin{array}{ccc} \text{Alg}_{\text{cat}}(\mathcal{V}) & \longrightarrow & \text{Alg}(\mathcal{V}) \\ \downarrow \lrcorner & & \downarrow \\ \text{Spc} & \xrightarrow{\text{Ass}_{(-)}} & \text{Op}^{ns} \end{array}$$

Moreover, $\text{Alg}_{\text{cat}}(\mathcal{V}) \rightarrow \text{Spc}$ is a Cartesian fibration, and given $f : A \rightarrow B \in \text{Alg}_{\text{cat}}(\mathcal{V})$ over $g : S_A \rightarrow S_B \in \text{Spc}$, we say f is $\mathcal{V}$ -enriched **fully faithful embedding** if f is a Cartesian lifting of g , that is, for any $x, y \in S_A$, there is an equivalence between enriched Homs $\underline{\text{Hom}}_A(x, y) \cong \underline{\text{Hom}}_B(g(x), g(y)) \in \mathcal{V}$. Furthermore, we can assemble $\text{Alg}_{\text{cat}}(\mathcal{V})$ together for different monoidal categories $\mathcal{V}$, and get a bifibration:

$$\text{Alg}_{\text{cat}} \rightarrow \text{Spc} \times \text{Mon}$$

Univalent/Complete categorical algebra

Given a monoidal category $\mathcal{V}$, let $E_{\mathcal{V}}^n$ denote the categorical algebra in $\mathcal{V}$ with the underlying space objects as the discrete set $\{0, 1, \dots, n\}$:

$$\text{Ass}_{\{0,1,\dots,n\}} \rightarrow \Delta^{op} \rightarrow \mathcal{V}^{\otimes}$$

We can use $E_{\mathcal{V}}^n$ to test categorical algebras.

Example

For categorical algebra $A : \text{Ass}_S \rightarrow \mathcal{V}$ and $E_{\mathcal{V}}^0$:

$$\text{Map}_{\text{Alg}_{\text{cat}}(\mathcal{V})}(E_{\mathcal{V}}^0, A) \cong S$$

Example

For $E_{\mathcal{V}}^1$, $\text{Map}_{\text{Alg}_{\text{cat}}(\mathcal{V})}(E_{\mathcal{V}}^1, A)$ can be regarded as the space of equivalences in A .

Remark

The point in $\text{Map}_{\text{Alg}_{\text{cat}}(\mathcal{V})}(E_{\mathcal{V}}^1, A)$ can be understood in the following way:

Definition

Given a monoidal category $\mathcal{V}$, a categorical algebra $A \in \text{Alg}_{\text{cat}}(\mathcal{V})$ is complete if $\text{Map}_{\text{Alg}_{\text{cat}}(\mathcal{V})}(E_{\mathcal{V}}^0, A) \rightarrow \text{Map}_{\text{Alg}_{\text{cat}}(\mathcal{V})}(E_{\mathcal{V}}^1, A)$ is an equivalence.

Example

The 'point' $E_{\mathcal{V}}^0$ is complete iff $\text{Aut}_{\mathcal{V}}(1) \cong *$, for example, when $\mathcal{V}$ is a Cartesian monoidal category.

Corollary

$\text{Cat}_{\mathcal{V}}$ is the localization of $\text{Alg}_{\text{cat}}(\mathcal{V})$ with respect to the morphism $E_{\mathcal{V}}^1 \rightarrow E_{\mathcal{V}}^0$.

Comparison between n -fold Segal space model

By Haugseng, there is a comparison of $\mathrm{Alg}_{\mathrm{cat}}(\mathrm{Cat})$ and $\mathrm{Seg}_2(\mathrm{Spc})$:

$$\begin{array}{ccc} \mathrm{Alg}_{\mathrm{cat}}(\mathrm{Cat}) & \xrightarrow{\cong} & \mathrm{Seg}_2(\mathrm{Spc}) \\ & \searrow & \swarrow \\ & \mathrm{Spc} & \end{array}$$

as Cartesian fibrations over Spc . Moreover, it restricts to an equivalence

$$\mathrm{Cat}^{\mathrm{Cat}} \cong \mathrm{Cat}_2$$

And similarly, $\mathrm{Cat}^{\mathrm{Cat}_{n-1}} \cong \mathrm{Cat}_n$.

Warning

In general, for the Cartesian fibration $\mathrm{Alg}_{\mathrm{cat}}(\mathcal{V}) \rightarrow \mathrm{Spc}$, the Cartesian lifting of a complex categorical algebra may not be complete again. For example, E^0 is complete in Spc , but not E^1 .

For the Cartesian fibration $\text{Alg}_{\text{cat}}(\mathcal{V}) \rightarrow \text{Spc}$, if we restrict to the fiber over $*$ $\in \text{Spc}$, we get a non-full subcategory of algebras in $\mathcal{V}$ $\text{Alg}(\mathcal{V})$. Moreover, we have the following theorem:

Theorem

Let B denote the functor $\text{Alg}(\mathcal{V}) \rightarrow \text{Alg}_{\text{cat}}(\mathcal{V})_{E^0/} \rightarrow \text{Cat}_{\text{loc}(E^0)/}^{\mathcal{V}}$, then B admits a right adjoint Ω .

A pointed map $\text{loc}(E^0) \rightarrow A$ in $\text{Cat}^{\mathcal{V}}$ is equivalent to a map $E^0 \rightarrow A$ in $\text{Alg}_{\text{cat}}(\mathcal{V})$, which is equivalent to the underlying space of A , thus given any point a in the underlying space of A , Ω sends it to the endomorphism object $A(a, a) \in \text{Alg}(\mathcal{V})$.

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Weakly left tensored category: another model

For a left tensored category $(\mathcal{V}, M)$, if the action map

$$act : \mathcal{V} \times M \rightarrow M$$

is closed, that is, for any $m \in M$, the induced map $act_m : \mathcal{V} \rightarrow M$ admits a right adjoint $\underline{\text{Hom}}_M(m, -) : M \rightarrow \mathcal{V}$, then M can be viewed as 'enriched' in $\mathcal{V}$.

Example

Cat as a left tensored category over itself, then the $act = \times : \text{Cat} \times \text{Cat} \rightarrow \text{Cat}$, given $\mathcal{C} \in \text{Cat}$, the right adjoint to $act_{\mathcal{C}}$ is $\text{Fun}(\mathcal{C}, -)$. So in this sense, Cat has mapping 'category' in the form of $\text{Fun}(\mathcal{C}, \mathcal{D})$. This should enable us to enhance Cat into an object in $\widehat{\text{Cat}}_2$.

Rigorous formulation of weakly left tensored category

One way to encode the left tensored category is by a morphism between coCartesian fibrations over Δ^{op}

$$M^{\otimes} \rightarrow \mathcal{V}^{\otimes}$$

such that, its fiber over $[0] \in \Delta^{op}$ is the map $M \rightarrow *$ encoding the underlying category acted by $\mathcal{V}$, and its fiber over $[1]$ is the projection $M \times \mathcal{V} \rightarrow \mathcal{V}$. The action map is encoded as the coCartesian liftings over the active map in $act : \{0\} : [0] \rightarrow [1] \in \Delta$, which gives rise to a functor $M \times \mathcal{V} \rightarrow M$.

Fix the notation: we call a morphism α in Δ^{op} preserves the maximum/minimum if its associated map in Δ preserves maximum/minimum.

Example

As we shall see, the map $\{0\} : [0] \rightarrow [n]$, preserving the minimum plays the role of 'active' map, and the map in the form of $\{n\} : [0] \rightarrow [n]$ preserving the maximum, plays the role of 'inert' maps.

Definition

Let $\mathcal{V}^\otimes \rightarrow \text{Ass}$ be a (non-symmetric) operad and $\phi : M^\otimes \rightarrow \mathcal{V}^\otimes$ a map of coCartesian fibrations relative to the inert maps of Ass that preserve the maximum.

We call $\phi : M^\otimes \rightarrow \mathcal{V}^\otimes$ a category weakly left tensored over $\mathcal{V}$ (or a category with weak left $\mathcal{V}$ -action) if the following conditions hold:

- 1 for every $n \geq 0$ the map $[0] \cong \{n\} \subset [n]$ in Δ induces an equivalence

$$\theta : M_{[n]}^\otimes \rightarrow \mathcal{V}_{[n]}^\otimes \times M_{[0]}^\otimes$$

- 2 for every $X, Y \in M^\otimes$ lying over $[m], [n] \in \text{Ass}$ the coCartesian lift $Y \rightarrow Y'$ of the map $[0] \cong \{n\} \subset [n]$ in Δ induces a pullback square:

$$\begin{array}{ccc} M^\otimes(X, Y) & \longrightarrow & M^\otimes(X, Y') \\ \downarrow & & \downarrow \\ \mathcal{V}^\otimes(\phi(X), \phi(Y)) & \longrightarrow & \mathcal{V}^\otimes(\phi(X), \phi(Y')) \end{array}$$

Let $\text{Cat}_{/\text{Ass}}^{\max} \subset \text{Cat}_{/\text{Ass}}$ be the subcategory of coCartesian fibrations relative to the collection of inert morphisms which preserves the maximum. Let

$$\omega\text{LMod} \subset \text{Op} \times_{\text{Fun}(\{1\}, \text{Cat}_{/\text{Ass}}^{\max})} \text{Fun}([1], \text{Cat}_{/\text{Ass}}^{\max})$$

be the full subcategories of categories with weak left action over some operad. Let

$$\text{LMod} \subset \text{Cat}_{/\text{Ass}}^{\text{coCart}} \times_{\text{Cat}_{/\text{Ass}}^{\text{coCart}}} \text{Fun}([1], \text{Cat}_{/\text{Ass}}^{\text{coCart}})$$

be the full subcategory of categories with left action of a monoidal category.

Theorem

(Heine) $\omega\text{LMod} \rightarrow \text{Op}$ is a biCartesian fibration, $\text{LMod} \rightarrow \text{Mon}$ is a Cartesian fibration, moreover, $\text{LMod} \rightarrow \omega\text{LMod}$ preserves Cartesian edges.

Definition

Let $\phi : M^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ be a weakly left tensored ∞ -category.

- 1 We say that ϕ exhibits M as (left) pseudo-enriched in $\mathcal{V}$ if $\mathcal{V}^{\otimes} \rightarrow \text{Ass}$ is a monoidal ∞ -category and for every $X, Y \in M$ and $V_1, \dots, V_n \in \mathcal{V}$, the map

$$\text{Mul}_M(V_1 \otimes \dots \otimes V_n, X; Y) \rightarrow \text{Mul}_M(V_1, \dots, V_n, X; Y)$$

induced by the active morphism $(V_1, \dots, V_n) \rightarrow V_1 \otimes \dots \otimes V_n$ is an equivalence.

- 2 a morphism object of $X, Y \in M$ is an object $\text{Mor}_M(X, Y) \in \mathcal{V}$ together with a multimorphism $\alpha \in \text{Mul}_M(\text{Mor}_M(X, Y), X; Y)$ that induces for all objects $Z_1, \dots, Z_n \in \mathcal{V}$ an equivalence of spaces:

$$\text{Mul}_{\mathcal{V}}(Z_1, \dots, Z_n; \text{Mor}_M(X, Y)) \cong \text{Mul}_M(Z_1, \dots, Z_n, X; Y)$$

is an equivalence.

Definition

We say that $\phi : M^{\otimes} \rightarrow \mathcal{V}^{\otimes}$ exhibits M enriched in $\mathcal{V}$ if for every $X, Y \in M$, there exists a morphism object $\text{Mor}_M(X, Y)$.

- 1 Let PLMod denote the non-full subcategory of ωLMod consisting of

pseudo-enriched categories and lax linear functors

$$\begin{array}{ccc} M^{\otimes} & \xrightarrow{f} & N^{\otimes} \\ \downarrow & & \downarrow \\ \mathcal{V}^{\otimes} & \xrightarrow{\alpha} & \mathcal{W}^{\otimes} \end{array}$$

where α is a monoidal functor.

- 2 Let ωPLMod be the full subcategory of ωLMod consisting of pseudo-enriched categories. Noticing that $\text{PLMod} \rightarrow \text{Mon}$ is the pullback of ωPLMod along the inclusion $\text{Mon} \rightarrow \text{Mon}^{\text{lax}}$ (where Mon^{lax} is the full subcategory of Op consisting of monoidal categories).
- 3 Let Enr denote the full subcategory of ωPLMod consisting of enriched categories.
- 4 Fix $\mathcal{V} \in \text{Op}$, let $\text{Enr}_{\mathcal{V}}$ denote the fiber of $\text{Enr} \rightarrow \text{Op}$ over $\mathcal{V}$.
- 5 Fix $\mathcal{V} \in \text{Mon}$, let $\text{PLMod}_{\mathcal{V}}$ denote the fiber of $\text{PLMod} \rightarrow \text{Mon}$ over $\mathcal{V}$.

Comparison

By Heine, we have comparisons:

- A fully faithful embedding $\text{Enr} \rightarrow \text{Alg}_{\text{cat}}$ consisting of complete categorical algebras.
- An equivalence $\text{Enr}_{\mathcal{V}} \cong \text{Cat}^{\mathcal{V}}$ for operad $\mathcal{V}^{\otimes}$.
- An equivalence $\text{PLMod}_{\mathcal{V}} \cong \text{Cat}^{\mathcal{P}(\mathcal{V})}$.
- An equivalence $\omega\text{LMod}_{\mathcal{V}} \cong \text{Cat}^{\mathcal{P}(\text{Env}(\mathcal{V}))}$.

Moreover, we have:

Theorem

(Heine)

- $\omega\text{LMod} \rightarrow \text{Op}$ is a biCartesian fibration
- $\text{PLMod} \rightarrow \text{Mon}$ is a biCartesian fibration.
- $\text{Enr} \rightarrow \text{Op}$ is a coCartesian fibration.
- The inclusion $\text{LMod} \rightarrow \text{PLMod}$ preserves Cartesian edges over Mon .
- The inclusion $\text{Enr} \rightarrow \text{PLMod}$ preserves coCartesian fibrations.

Adjunction

Without abuse of notation, we also use $\text{Enr} \rightarrow \text{Mon}$ to denote the pullback of $\text{Enr} \times_{\text{Op}} \text{Mon}$. Let $P\text{Enr}$ denote the pullback of $\text{Enr} \rightarrow \widehat{\text{Mon}}$ along $\mathcal{P}(-) : \text{Mon} \rightarrow \widehat{\text{Mon}}$. Then we have:

Theorem

There exists an equivalence $P\text{Enr} \cong \text{PLMod}$ over Mon . Fiberwise, it sends a $\mathcal{P}(\mathcal{V})$ enriched category $M^{\otimes} \rightarrow \mathcal{P}(\mathcal{V})^{\otimes}$ to pullback along $\mathcal{V} \hookrightarrow \mathcal{P}(\mathcal{V})$, which is a pseudo-enriched category in $\mathcal{V}$.

Theorem

(Heine) Given a monoidal adjunction $f : \mathcal{V}^{\otimes} \rightleftharpoons \mathcal{W}^{\otimes}$, then there is an induced adjunction:

$$f_! : \text{Enr}_{\mathcal{V}} \rightleftharpoons \text{Enr}_{\mathcal{W}} : g_!$$

Thus, given a morphism $\alpha : \mathcal{V} \rightarrow \mathcal{W}$ in Mon , the adjunction classified by the biCartesian fibration $\text{PLMod} \rightarrow \text{Mon}$ is equivalent to the transfer of enrichment along the adjunction $\alpha_! : \mathcal{P}(\mathcal{V}) \rightleftharpoons \mathcal{P}(\mathcal{W}) : \alpha^*$.

Symmetric monoidal structure

Consider the Cartesian symmetric monoidal structure on $\mathbf{Enr}^\times$; then the coCartesian fibration $\mathbf{Enr} \rightarrow \mathbf{Mon}^{lax}$ can be enhanced into a coCartesian fibration $\mathbf{Enr}^\times \rightarrow (\mathbf{Mon}^{lax})^\times$. Given any $\mathcal{O}^\otimes$ algebra object $\mathcal{V}$ in $\mathbf{Mon}^{lax} \cong \mathbf{Alg}(\mathbf{Cat})$, which is equivalent to an object in $\mathbf{Alg}_{\mathbb{E}_1 \otimes \mathcal{O}}(\mathbf{Cat})$, it gives rise to a symmetric monoidal category

$$\begin{array}{ccc} (\mathbf{Cat}^\mathcal{V})^\otimes & \longrightarrow & \mathbf{Enr}^\times \\ \downarrow \lrcorner & & \downarrow \\ \mathbf{Comm}^\otimes & \longrightarrow & \mathbf{Mon}^\times \end{array}$$

Similarly, given a morphism $f : \mathcal{V} \rightarrow \mathcal{W}$ between $\mathcal{O}$ algebra objects in $\mathbf{Mon}$, the transfer of enrichment $f_! : \mathbf{Cat}^\mathcal{V} \rightarrow \mathbf{Cat}^\mathcal{W}$ can be enhanced into a symmetric monoidal functor:

$$(\mathbf{Cat}^\mathcal{V})^\otimes \rightarrow (\mathbf{Cat}^\mathcal{W})^\otimes$$

Symmetric monoidal Version of the results we have already known

Adjunction: Given a symmetric monoidal adjunction $f : \mathcal{V}^\otimes \rightleftarrows \mathcal{W}^\otimes : g$, we get a symmetric monoidal functor $f_!^\otimes : (\text{Cat}^\mathcal{V})^\otimes \rightarrow (\text{Cat}^\mathcal{W})^\otimes$, thus the right adjoint $g_!$ can be enhanced into a lax symmetric monoidal functor, and we get a symmetric monoidal adjunction:

$$f_!^\otimes : (\text{Cat}^\mathcal{V})^\otimes \rightleftarrows (\text{Cat}^\mathcal{W})^\otimes : g_!^\otimes$$

Equivalences : Recall the equivalences $\text{PLMod}_\mathcal{V} \cong \text{Cat}^\mathcal{V}$, if $\mathcal{V}$ is symmetric monoidal, then likewise, we can equip $\text{PLMod}_\mathcal{V}$ with a symmetric monoidal structure by the pullback

$$\begin{array}{ccc} (\text{Cat}^\mathcal{V})^\otimes & \longrightarrow & \text{PLMod}^\times \\ \downarrow \lrcorner & & \downarrow \\ \text{Comm}^\otimes & \longrightarrow & \text{Mon}^\times \end{array}$$

And we can enhance the equivalence above into $(\text{Cat}^\mathcal{V})^\otimes \cong (\text{PLMod}_\mathcal{V})^\otimes$

De-looping: Now we can make the de-looping adjunction functorial:

$B : \text{Alg}^\times \rightarrow \text{Alg}_{\text{cat}}^\times \xrightarrow{\text{Loc}} \text{Enr}^\times$ over $\text{Mon}^\times$, thus given a symmetric monoidal category $\mathcal{V}$, by pullback, we get a symmetric monoidal functor:

$$B_{\mathcal{V}} : \text{Alg}(\mathcal{V})^\otimes \rightarrow (\text{Cat}^{\mathcal{V}})^\otimes$$

, thus the de-looping adjunction can be enhanced into a symmetric monoidal adjunction

$$B_{\mathcal{V}} : \text{Alg}(\mathcal{V})^\otimes \rightleftarrows (\text{Cat}^{\mathcal{V}})^\otimes : \Omega$$

$$\mathrm{LMod}_{\mathcal{V}}(\mathrm{Cat})^{\otimes} \rightarrow (\mathrm{Cat}^{\mathcal{P}(\mathcal{V})})^{\otimes}$$

Consider the symmetric monoidal inclusion $\mathrm{LMod}^{\times} \rightarrow \mathrm{PLMod}^{\times}$ over $\mathrm{Mon}^{\times}$, given a symmetric monoidal category $\mathcal{V}^{\otimes}$, specified by a commutative algebra object $\mathrm{Comm}^{\otimes} \rightarrow \mathrm{Mon}^{\times}$, then by pullback along it, we get a lax symmetric monoidal functor

$$\mathrm{LMod}_{\mathcal{V}}(\mathrm{Cat})^{\otimes} \rightarrow (\mathrm{Cat}^{\mathcal{P}(\mathcal{V})})^{\otimes}$$

Moreover, since $\mathrm{LMod} \rightarrow \mathrm{PLMod}$ preserves Cartesian edges, for any symmetric monoidal functor $\mathcal{V} \rightarrow \mathcal{W}$, there is:

$$\begin{array}{ccc} \mathrm{LMod}_{\mathcal{W}}^{\otimes} & \longrightarrow & \mathrm{PLMod}_{\mathcal{W}}^{\otimes} \\ \downarrow f! & & \downarrow f^* \\ \mathrm{LMod}_{\mathcal{V}}^{\otimes} & \longrightarrow & \mathrm{PLMod}_{\mathcal{V}}^{\otimes} \end{array}$$

Example

Let $\mathcal{V} = *$, then the lax symmetric monoidal functor becomes symmetric monoidal equivalences of $\mathrm{Cat}^{\times}$.

Symmetric monoidal enriched Yoneda embedding

Consider the ‘large’ version of categories enriched in a presentable category $\mathcal{V}$, then any presentably left tensored module $(\mathcal{V}, M) \in \mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L)$ is closed, thus is enriched in $\mathcal{V}$, especially, we have an inclusion:

$$\mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L) \rightarrow \widehat{\mathbf{Cat}}^{\mathcal{V}}$$

Moreover, by Heine, the inclusion admits a symmetric monoidal left adjoint, which restricts to the enriched Yoneda embedding on the full subcategory $\mathbf{Cat}^{\mathcal{V}} \hookrightarrow \widehat{\mathbf{Cat}}^{\mathcal{V}}$:

$$\mathcal{P}_{\mathcal{V}}(-) : \mathbf{Cat}^{\mathcal{V}} \rightarrow \mathbf{LMod}_{\mathcal{V}}(\mathbf{Pr}^L)$$

, where the unit map of the symmetric monoidal adjunction above valued at $\mathcal{C} \in \mathbf{Cat}^{\mathcal{V}}$ gives us the enriched fully faithful ‘symmetric monoidal’ Yoneda embedding:

$$\mathcal{C} \hookrightarrow \mathcal{P}_{\mathcal{V}}(\mathcal{C}) \cong \mathbf{Fun}^{\mathcal{V}}(\mathcal{C}^{op}, \mathcal{V}) \in \mathbf{CAlg}(\mathbf{Cat}^{\mathcal{V}})$$

Especially, let $\mathcal{V} = \mathbf{Cat}$, then the $\mathbf{Cat}$ enriched Yoneda embedding is a symmetric monoidal $(\infty, 2)$ functor.

$$\text{LMod}_{\text{Cat}}(\text{Pr}^L) \rightarrow \widehat{\text{Cat}}_2$$

By the inclusion $\text{LMod}_{\text{Cat}}(\text{Pr}^L)^{\otimes} \rightarrow \widehat{\text{Cat}}_2^{\times}$, we can enhance any object $\mathcal{C} \in \text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^L))$ to a symmetric monoidal $(\infty, 2)$ category:

Example

Let $\text{Spc} \rightarrow \text{Cat}$ be the symmetric monoidal embedding, then by transfer of enrichment, there is a (fully faithful) symmetric monoidal embedding $\text{Cat}^{\text{Spc}} \cong \text{Cat}^{\times} \rightarrow \text{Cat}_2^{\times}$, which provides us with a symmetric monoidal $(\infty, 2)$ structure on Cat_2 .

Thus we can talk about adjunction in Cat_2 , especially the 2-morphism is in the form of a 2 functor $[1] \times C \rightarrow D$, thus agrees with the one we defined earlier.

Theorem

(Heine/Shay Ben-Moshe) The embedding $\text{LMod}_{\text{Cat}}(\widehat{\text{Cat}})_{\text{closed}}^{\text{lax}} \rightarrow \widehat{\text{Cat}}_2$ can be enhanced into a 2 functor, thus preserving adjunctions.

The 2 category structure on $\mathbf{LMod}_{\mathbf{Cat}}(\widehat{\mathbf{Cat}})^{lax}_{closed}$ inherits from the 2 category structure on $\widehat{\mathbf{Cat}}_{/\mathcal{LM}^{\otimes}}$, especially, the adjunction in $\mathbf{LMod}_{\mathbf{Cat}}(\widehat{\mathbf{Cat}})^{lax}_{closed}$ is relative adjunction over the operad $\mathcal{LM}^{\otimes}$.

Fact

Given a left module morphism $F : (A, M) \rightarrow (A, N)$, if F admits a right adjoint F^R at the underlying category level, then it can be enhanced into an adjunction in $\mathbf{LMod}_A(\widehat{\mathbf{Cat}})^{lax}$.

Now we have a way to produce lax symmetric monoidal $(\infty, 2)$ functors:

Corollary

Given a morphism $F : C \rightarrow D$ in $\mathbf{CAlg}(\mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L)) \rightarrow \mathbf{CAlg}(\mathbf{Cat}_2)$, if the underlying functor of F admits a right adjoint G , then G can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor.

Remark

A morphism F in $\mathbf{CAlg}(\mathbf{LMod}_{\mathbf{Cat}}(\mathbf{Pr}^L)) \cong \mathbf{CAlg}(\mathbf{Pr}^L)_{\mathbf{Cat}} /$ corresponds to a diagram $\mathbf{Cat} \rightarrow C \rightarrow D$ in $\mathbf{Pr}^L$.

- 1 Main conclusion
- 2 $\text{Span}_n(\mathcal{C})$
- 3 Symmetric monoidal (∞, n) structure
- 4 Enriched ∞ -category theory
- 5 Another model of enriched category theory
- 6 Enhance $\text{Span}_n(\mathcal{C})$ into $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}(n \text{ Cat}^{\text{Span}_1(\mathcal{C})})$**
- 7 A lax symmetric monoidal $(\infty, 2)$ functor $\text{Span}_2(\mathcal{C}) \rightarrow 2\text{Pr}_\kappa$
- 8 Applications

$$\text{Span}_2(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\text{Cat}^{\text{Span}_1(\mathcal{C})}})$$

Construction

Let $\mathcal{C}$ be a category with finite limits.

- Input $\text{Span}_2(\mathcal{C}) \in \text{CAlg}(\text{Cat}_2 \cong \text{Cat}^{\text{Cat}})$, valued by the unit of the symmetric monoidal de-looping adjunction, we get an $(\infty, 2)$ fully faithful embedding $B_{\text{Cat}}\Omega(\text{Span}_2(\mathcal{C})) \hookrightarrow \text{Span}_2(\mathcal{C})$.
- We can compute that the underlying $(\infty, 1)$ category of $B_{\text{Cat}}\Omega(\text{Span}_2(\mathcal{C})) \cong B \text{Span}_1(\mathcal{C})$, moreover, we can identify the symmetric monoidal structure induced by the symmetric monoidal adjunction with the one acquired by functoriality $\text{Span}_1(-)$ and then compose with B_{Cat} .
- Now input $B \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\text{Cat}})$, this is equivalent to enhancing $\text{Span}_2(\mathcal{C})$ into an object in $\text{CAlg}(\text{LMod}_{B \text{Span}_1(\mathcal{C})}(\text{Cat}^{\text{Cat}}))$.

Construction

- Given $A \in \text{CAlg}(\mathcal{M})$ for $\mathcal{M} \in \text{CAlg}(\text{Pr}^L)$, consider the symmetric monoidal adjunction

$$(-) \otimes A : \mathcal{M} \rightleftharpoons \text{LMod}_A(\mathcal{M}) : \text{forget}$$

, there is an induced symmetric monoidal equivalence

$$\text{Cat}^{\text{LMod}_A(\mathcal{M})} \cong \text{LMod}_{B_{\mathcal{M}}A}(\text{Cat}^{\mathcal{M}})$$

- Thus we get an object in $\text{CAlg}(\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})})$ from the input data $B_{\text{Cat}} \text{Span}_1(\mathcal{C}) \rightarrow \text{Span}_2(\mathcal{C})$, moreover, one can compute the enriched Hom between $x, y \in \text{Span}_2(\mathcal{C})$: $\underline{\text{Hom}}(x, y) = \text{Span}_1(\mathcal{C}_{/x \times y})$ under the $\text{Span}_1(\mathcal{C})$ action by taking product, which is closed.
- Thus as an object, $\text{Span}_2(\mathcal{C})$ actually lands inside the full subcategory $\text{Cat}^{\text{LMod}_{\text{Span}_1(\mathcal{C})}(\text{Cat})_{\text{closed}}}$.

Construction

- The lax symmetric monoidal inclusion $LMod_{Span_1(\mathcal{C})}(\mathcal{Cat}) \rightarrow \mathcal{Cat}^{\mathcal{P}(Span_1(\mathcal{C}))}$ factors through the full symmetric monoidal subcategory $\mathcal{Cat}^{Span_1(\mathcal{C})} \hookrightarrow \mathcal{Cat}^{\mathcal{P}(Span_1(\mathcal{C}))}$, thus we get a lax symmetric monoidal functor:

$$(\mathcal{Cat}^{LMod_{Span_1(\mathcal{C})}(\mathcal{Cat})})^{\otimes} \rightarrow (\mathcal{Cat}^{\mathcal{Cat}^{Span_1(\mathcal{C})}})^{\otimes} = 2(\mathcal{Cat}^{Span_1(\mathcal{C})})^{\otimes}$$

- And thus we enhance $Span_2(\mathcal{C})$ into an object in $CAlg(2 \mathcal{Cat}^{Span_1(\mathcal{C})})$, denoted as $Span_2^{enr}(\mathcal{C})$.
- Inductively, for $n > 2$, we can enhance $Span_n(\mathcal{C})$ into an object in $CAlg(n \mathcal{Cat}^{Span_1(\mathcal{C})})$, and we can recover the symmetric monoidal (∞, n) structure on $Span_n(\mathcal{C})$ by transfer of enrichment along the lax symmetric monoidal functor $\Gamma : \mathcal{Cat}^{Span_1(\mathcal{C})} \rightarrow \mathcal{Cat}$.

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- 5 Another model of enriched category theory
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- 7 A lax symmetric monoidal $(\infty, 2)$ functor $\text{Span}_2(\mathcal{C}) \rightarrow 2\text{Pr}_\kappa$
- 8 Applications

$$\text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$$

Now we can construct a ‘six functor formalism’, which is a lax symmetric monoidal $(\infty, 2)$ functor

$$\text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{\text{Span}_1(\mathcal{C})}$$

sending $x \in \mathcal{C}$ to $\text{Span}_1(\mathcal{C}_{/x})$, enriched in $\text{Span}_1(\mathcal{C})$ under the action:

$$\text{act} : \text{Span}_1(\mathcal{C}) \times \text{Span}_1(\mathcal{C}_{/x}) \rightarrow \text{Span}_1(\mathcal{C}_{/x}), (c, x) \mapsto c \times x$$

The symmetric monoidal $(\infty, 2)$ structure on $\text{Cat}^{\text{Span}_1(\mathcal{C})}$ inherits from the symmetric monoidal $(\infty, 2)$ structure on $\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$:

$$\text{Cat}^{\mathcal{P}(\ast)} \cong \text{Cat} \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} \in \text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^L))$$

.

Consider $\text{Span}_2^{\text{enr}}(\mathcal{C}) \in \text{CAlg}(\text{Cat}^{\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}})$, let $\mathcal{V}$ denote $\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$, and consider the enriched Yoneda embedding:

$$\text{Span}_2^{\text{enr}}(\mathcal{C}) \hookrightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_2^{\text{enr}}(\mathcal{C}))$$

in $\text{CAlg}(\widehat{\text{Cat}}^{\mathcal{V}})$. On the other hand, there is map in $\text{CAlg}(\text{LMod}_{\mathcal{V}}(\text{Pr}^L))$:

$$1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\text{Span}_2^{\text{enr}}(\mathcal{C})), 1_{\mathcal{V}} \mapsto 1_{\mathcal{P}_{\mathcal{V}}(\text{Span}_2^{\text{enr}}(\mathcal{C}))}$$

, moreover, it admits a right adjoint inside $\text{LMod}_{\mathcal{V}}(\text{Pr}^L) \rightarrow \widehat{\text{Cat}}^{\mathcal{V}}$, denoted as $\Gamma : \mathcal{P}_{\mathcal{V}}(\text{Span}_2^{\text{enr}}(\mathcal{C})) \rightarrow \mathcal{V}$, sending $x \in \text{Span}_2^{\text{enr}}(\mathcal{C})$ to $\underline{\text{Hom}}_{\mathcal{P}_{\mathcal{V}}(\text{Span}_2^{\text{enr}}(\mathcal{C}))}(1, x) \cong \text{Span}_1^{\text{enr}}(\mathcal{C}_{/x})$.

Corollary

we have a chain of morphisms in $\text{CAlg}(\widehat{\text{Cat}}^{\mathcal{V}})$:

$$\text{Span}_n^{\text{enr}}(\mathcal{C}) \xrightarrow{y_{\mathcal{V}}} \mathcal{P}_{\mathcal{V}}(\text{Span}_n^{\text{enr}}(\mathcal{C})) \xleftarrow{1} \mathcal{V}$$

where 1 admits a right adjoint Γ .

Consider the canonical symmetric adjunction

$$F : \mathbf{Spc} \rightleftarrows \mathcal{P}(\mathbf{Span}_1(\mathcal{C})) : G$$

where F is the unique symmetric monoidal colimit-preserving functor from $\mathbf{Spc}$ to $\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))$. Then there exists a symmetric monoidal adjunction

$$F_{2,*} : \mathbf{Cat}_2 \rightleftarrows 2\mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} : G_{2,*}$$

especially, the right adjoint $G_* : (n-1)\mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))} \rightarrow \mathbf{Cat}_{n-1}$ can be enhanced into a lax symmetric monoidal (∞, n) functor.

Corollary

Under the lax symmetric monoidal $(\infty, 2)$ functor $G_{2,} : \mathbf{CAlg}(2\mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}) = \mathbf{CAlg}(\mathbf{Cat}^{\mathcal{V}}) \rightarrow \mathbf{CAlg}(\mathbf{Cat}_2)$, there are morphisms:*

$$\mathbf{Span}_2(\mathcal{C}) \xrightarrow{y_{\mathcal{V}}} \mathcal{P}_{\mathcal{V}}(\mathbf{Span}_2^{\text{enr}}(\mathcal{C})) \xleftarrow{1} \mathcal{V}$$

where $y_{\mathcal{V}}$ is a symmetric monoidal $(\infty, 2)$ functor and $1 : \mathcal{V} \rightarrow \mathcal{P}_{\mathcal{V}}(\mathbf{Span}_2^{\text{enr}}(\mathcal{C}))$ admits the right adjoint Γ in $\mathbf{Cat}_2$ since $G_{2,}$ preserves adjunction. Thus Γ can be enhanced into a lax symmetric monoidal $(\infty, 2)$ functor, so is $\Gamma \circ y_{\mathcal{V}} : \mathbf{Span}_2(\mathcal{C}) \rightarrow \mathbf{Cat}^{\mathcal{P}(\mathbf{Span}_1(\mathcal{C}))}$.*

Remark on the symmetric monoidal $(\infty, 2)$ structure on $\text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$

Noticing that under the lax symmetric monoidal functor $G_{2,*}$ above, $\mathcal{V} \in \text{CAlg}(\text{LMod}_{\mathcal{V}}(\text{Pr}^L))$ recovers the symmetric monoidal $(\infty, 2)$ structure on $\mathcal{V} = \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))}$ through the symmetric monoidal colimit-preserving functor $\text{Cat} \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} \in \text{CAlg}(\text{LMod}_{\text{Cat}}(\text{Pr}^L)) \rightarrow \text{CAlg}(\text{Cat}_2)$, by the following diagram:

$$\begin{array}{ccc} \text{LMod}_{\mathcal{V}}(\text{Pr}^L) & \longrightarrow & \widehat{\text{Cat}}^{\mathcal{V}} \\ \text{forget} \downarrow & & \downarrow G_{2,*} \\ \text{LMod}_{\text{Cat}}(\text{Pr}^L) & \longrightarrow & \widehat{\text{Cat}}_2 \end{array}$$

Given the lax symmetric monoidal $(\infty, 2)$ functor

$$\mathrm{Span}_2(\mathcal{C}) \rightarrow \mathrm{Cat}^{\mathrm{Span}_1(\mathcal{C})} \hookrightarrow \mathrm{Cat}^{\mathcal{P}(\mathrm{Span}_1(\mathcal{C}))}$$

, and a sheaf theory:

$$\mathrm{Shv}(-) : \mathrm{Span}_1(\mathcal{C}) \rightarrow \mathrm{Pr}^L$$

, assume that our sheaf theory lands inside Pr_κ for some regular cardinality, then we have an induced symmetric monoidal colimit preserving functor in Pr^L :

$$\mathcal{P}(\mathrm{Span}_1(\mathcal{C})) \rightarrow \mathrm{Pr}_\kappa^L$$

over $\mathrm{Spc} \cong \mathcal{P}(\ast)$, thus we get a commutative diagram:

$$\begin{array}{ccc} \mathrm{Cat}^{\mathcal{P}(\mathrm{Span}_1(\mathcal{C}))} & \xrightarrow{\quad} & \mathrm{Cat}^{\mathrm{Pr}_\kappa} \\ & \nwarrow \quad \nearrow & \\ & \mathrm{Spc} & \end{array}$$

. Thus, we get a symmetric monoidal

$(\infty, 2)$ functor $\mathrm{Cat}^{\mathcal{P}(\mathrm{Span}_1(\mathcal{C}))} \rightarrow \mathrm{Cat}^{\mathrm{Pr}_\kappa}$.

$$\text{Cat}^{\text{Pr}_\kappa} \rightarrow 2\text{Pr}_\kappa$$

Notice the following commutative diagram:

$$\begin{array}{ccccc}
 \text{Cat}^{\text{Pr}_\kappa} & \xrightarrow{\mathcal{P}_{\text{Pr}_\kappa}(-)} & \text{LMod}_{\text{Pr}_\kappa}(\text{Pr}_\kappa) & \longrightarrow & \text{LMod}_{\text{Pr}_\kappa}(\text{Pr}^L) \\
 \uparrow (\text{Spc} \rightarrow \text{Pr}_\kappa)_* & & \uparrow (-) \otimes \text{Pr}_\kappa & & \uparrow (-) \otimes \text{Pr}_\kappa \\
 \text{Cat}^{\text{Spc}} \cong \text{Cat} & \xrightarrow{\mathcal{P}(-)} & \text{Pr}_\kappa & \longrightarrow & \text{Pr}^L
 \end{array}$$

Thus we get a symmetric monoidal $(\infty, 2)$ functor

$$\text{Cat}^{\text{Pr}_\kappa} \rightarrow 2\text{Pr}_\kappa$$

And by composition of all the functors we have acquired:

Corollary

There exists a lax symmetric monoidal $(\infty, 2)$ functor

$$\text{Span}_2(\mathcal{C}) \rightarrow \text{Cat}^{\mathcal{P}(\text{Span}_1(\mathcal{C}))} \rightarrow \text{Cat}^{\text{Pr}_\kappa} \rightarrow 2\text{Pr}_\kappa$$

- 1 Main conclusion
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- 6 Enhance $\text{Span}_n(\mathcal{C})$ into $\text{Span}_n^{\text{enr}}(\mathcal{C}) \in \text{CAlg}(n \text{ Cat}^{\text{Span}_1(\mathcal{C})})$
- 7 A lax symmetric monoidal $(\infty, 2)$ functor $\text{Span}_2(\mathcal{C}) \rightarrow 2\text{Pr}_{\kappa}$
- 8 Applications**

Why lax symmetric monoidal $(\infty, 2)$?

There is an $(\infty, 2)$ -operad theory working on progress by Haugseng's student Tallak Manum, an interesting example of such an $(\infty, 2)$ operad is $\mathbb{E}_2^{lax}$, which is the gray-BV tensor product of $\mathbb{E}^1$ with itself (recall that the BV-tensor product of $\mathbb{E}_1$ with itself is $(\infty, 1)$ operad $\mathbb{E}_2$). Especially, a lax symmetric monoidal $(\infty, 2)$ functor preserves $\mathbb{E}_2^{lax}$ algebra. And we can use this to construct $\mathbb{E}_2^{lax}$ algebra object. In fact, $\mathbb{E}_2^{lax}$ algebra in $\text{Span}_2(\mathcal{C})$ can be nicely described:

Theorem

(Ho-Walker-Jonte):

$$\text{Alg}_{\mathbb{E}_2^{lax}}(\text{Span}_2(\mathcal{C}))^{\simeq} \cong 2\text{Seg}(\mathcal{C})^{\simeq} \subset \text{Fun}(\Delta^{op} \times \Delta^{op}, \mathcal{C})^{\simeq}$$

Thus, if we have a lax symmetric monoidal $(\infty, 2)$ functor $\text{Span}_2(\mathcal{C})^{\otimes} \rightarrow D^{\otimes}$, by constructing a 2-Segal object in $\mathcal{C}$, we can acquire a lax $\mathbb{E}_2$ object in $D^{\otimes}$.

2 Shv(—)

Higher sheaf and Higher algebraic geometry are recently developed by Germán Stefanich and Peter Scholze.

One can show that if we take our 1-sheaf theory: $\text{Shv}(—)$ to be $\text{QCoh} : \text{Span}_1((DRing)^{op}) \rightarrow 1\text{Pr}$, then $2\text{Shv}(—)$ is precisely $\text{Mod}_{\text{QCoh}(—)}(\text{DGCat}) = 2\text{QCoh}(—)$, and we can further right Kan extend from $DRing^{op}$ to schemes and stacks, this recovers the theory of 'sheaf of categories' developed by Gaitsgory.

Example

When X is highly stacky, higher sheaf provides more geometric information: $2\text{QCoh}(B^2G_m) \cong \lim_{X \in DRing^{op}/B^2G_m} 2\text{QCoh}(X)$ contains an object $(\text{LMod}_A(\text{QCoh}(X)) \in 2\text{QCoh}(X))_X$ where A is the Azuyama algebra associated to $[X] \in H^2(X, G_m)$. This is an object similar to the canonical line bundle on $\text{QCoh}(BG_m)$.

From now on, we fix a sheaf theory: Let $\mathcal{C} = \text{Stk}$, which is the category of Artin stacks of finite type with affine stabilizer over $\overline{k_q}$ where $q = p^n$ for a prime number p , and G is a smooth algebraic group over $\overline{k_q}$. our sheaf theory as the one developed by (Liu-Zheng), which is a lax symmetric monoidal functor:

$$\text{Shv}(-) : \text{Span}_1(\mathcal{C}) \rightarrow \text{Pr}^L$$

assigning a stack $X \in \text{Stk}$ with the ∞ -category of l -adic sheaves $\text{Shv}(X)$ on X , and given a span $x \xleftarrow{f} z \xrightarrow{g} y$ it sends f to $f^* : \text{Shv}(z) \rightarrow \text{Shv}(x)$, and g to $g_! : \text{Shv}(z) \rightarrow \text{Shv}(y)$.

The l -adic sheaf theory we take $\mathrm{Shv}(-) : \mathrm{Span}(\mathrm{Stk}) \rightarrow \mathrm{Pr}^L$ lands inside the non-full subcategory $1\mathrm{Pr} = \mathrm{Pr}_{\aleph_1}$:

- For $\mathcal{X} \in \mathrm{Stk}$ with atlas of a scheme $j : X \rightarrow \mathcal{X}$, $\mathrm{Shv}(\mathcal{X})$ is compactly generated by objects in the form of $j_!(\mathcal{F})$ for compact object $\mathcal{F} \in \mathrm{Shv}(X)^c$.
- For a morphism $f : \mathcal{X} \rightarrow \mathcal{Y}$ in Stk , $f_!$ preserves compactness since $f^!$ is continuous.
- However, f^* doesn't preserve compactness in general (for example, $\underline{1}_{BG_n}$ is not compact). Instead, f^* preserves constructibility, and a constructible sheaf on any $\mathcal{X} \in \mathcal{C}$ is $\aleph_1$ -compact. Thus, for an atlas $h : Y \rightarrow \mathcal{Y}$ and compact $\mathcal{F} \in \mathrm{Shv}(Y)^c$, we take the $\aleph_1$ -compact generator $h_!\mathcal{F}$ of $\mathcal{Y}$, and by base change, we have $f^*h_!(\mathcal{F}) \cong h'_!(f')^*(\mathcal{F})$, which is $\aleph_1$ -compact.

$$\begin{array}{ccc} Y \times_{\mathcal{Y}} \mathcal{X} & \xrightarrow{f'} & Y \\ h' \downarrow & & \downarrow h \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

$2\mathrm{Shv}(BG)$

Given a smooth algebraic group G , consider its classifying stack BG , which is the colimit of the groupoid object $G^\bullet$ inside Stk :

$$\mathrm{colim}(\cdots \rightarrow G^2 \begin{matrix} \rightarrow \\ \rightarrow \\ \rightarrow \end{matrix} G \begin{matrix} \rightarrow \\ \rightarrow \end{matrix} *) = BG$$

One can show that the following diagram is a limit diagram in $\mathrm{LMod}_{1\mathrm{Pr}}(1\mathrm{Pr})$:

$$2\mathrm{Shv}(BG) \xrightarrow{\cong} \mathrm{Tot}(2\mathrm{Shv}(*) \rightarrow 2\mathrm{Shv}(G) \rightrightarrows 2\mathrm{Shv}(G^2) \begin{matrix} \rightarrow \\ \rightarrow \\ \rightarrow \end{matrix} \cdots)$$

Corollary

The biadjunction

$$2\mathrm{Shv}(BG) \rightleftarrows 2\mathrm{Shv}(*)$$

induced by the atlas $$ $\rightarrow BG$ is monadic, the monad is the group scheme $G \in \mathrm{Stk}$, and there is equivalence $2\mathrm{Shv}(BG) \cong \mathrm{LMod}_G(2\mathrm{Shv}(*))$.*